Applied Optoelectronics | AAOI

Full investment research and earnings rescore | 5 November 2026 | Framework 8.0 | NASDAQ:AAOI | USD

Current investment view

AAOI is converting hyperscale optical demand into a faster 800G-to-1.6T product ramp. Its internal lasers, Texas manufacturing expansion and customer relationships create substantial potential operating leverage. The forecast depends on repeatable good output, customer acceptance, available DSPs, contracted prices and enough capital to fund the factories and receivables.

The active Q4 2027 calculated ceiling is $345. It represents the Q4 2027 through Q3 2028 earnings window under the established 20x modeled operating-profit method. The ceiling requires a large forecast earnings ramp and conditional financing; it provides a scenario valuation for the stated horizon.

The forecast follows the late-2027 monthly datacenter-transceiver milestone, the staged Texas build and a faster 1.6T product mix. Factory costs, contractual price erosion, working capital and dilution absorb part of the benefit from higher shipments.

Analytical Base remains 70/100, from 42/60 points. Protected Excel Base remains 72/100. The company method stays at 20x forward-four-quarter fully expensed modeled operating profit with starting-quarter economic shares. The score contributes no additional multiplier to a ceiling.

1. Results and management guidance

The Q3 release, financial statements and complete management call provide the current operating anchor. Q3 revenue of $270 million included $156 million datacenter, $111 million CATV, $3 million telecom and less than $1 million other revenue. Management reported $48 million of 800G revenue and $93 million of 400G revenue. The first 1.6T revenue is expected in Q4.

Q3 2026 actual, USD M	Reported
Revenue	270
GAAP gross profit / gross margin	76 / 28%
GAAP operating expenses	78
GAAP operating loss	(1)
GAAP net income	6
Non-GAAP operating profit, reconciled	7
Non-GAAP net income / diluted EPS	15 / $0.17
Basic / diluted weighted shares, M	85 / 88

The non-GAAP operating bridge adds back stock compensation, amortization, discontinued-business costs and designated nonrecurring items. Restoring recurring stock compensation and amortization produces about $1 million of recurring fully expensed Q3 operating profit. Government subsidies in other income remain outside operating profit. Historical displayed forecasts retain their as-published values alongside this separately dated actual-results anchor.

Q4 2026 guidance	Company	Current model
Total revenue, USD M	385-420	399
Datacenter revenue, USD M	270-300	286
CATV revenue, USD M	105-115	110
1.6T revenue, USD M	50-70	60
Non-GAAP gross margin	29.5%-31%	30%
Non-GAAP opex, USD M	68-75	72
Non-GAAP net income, USD M	45.9-60.4	Separate basis
Company diluted EPS / shares	0.48-0.63 / 96M	Separate basis

The model fits revenue, mix, gross-margin and opex guidance. Its fully expensed operating profit is $43 million and recurring net income is approximately $35 million. Company non-GAAP earnings guidance uses its own adjustment, tax and other-income assumptions. The model expenses recurring compensation and amortization, assumes normalized book tax of 21% and excludes one-time subsidies and any deferred-tax valuation-allowance release from valued operating profit. These basis differences explain the separate net-income presentation.

2. Current management guidance and customer contracts

The complete Q3 and Q2 prepared remarks and Q&A were read. Current management guidance provides the power, construction, component and customer-order sequence used in the operating model.

Operating item	Current management expectation	Model treatment
Year-end 2026 capacity	Above 500K/month	Quarter-average equipment capacity
Year-end 2027 capacity	Above 1M/month	Accepted output after mix / conversion
Texas fraction at end 2027	About 30%	US and overseas production retain their scopes
Monthly DC revenue milestone	$471M late / end 2027	Finite delivery ramp
Sugar Land phase 1	December 2026	Phased commissioning
2027 capital payments	CEO Q&A: demand / payment timing dependent	State-specific capital cash schedules

Equipment capacity currently approaches 330,000 modules per month, compared with roughly 200,000 in June. Phase 2 has faced power, transformer, cable and construction delays. The model treats factory installation, qualifying output and accepted shipments as separate steps. It uses quarter-average equipment capacity, product allocation and net realized ASPs to constrain recognized sales.

Management describes continuing demand above available production, an approximately linear ramp toward the monthly revenue milestone and enough prospective demand to support about 1.2 million monthly modules. The larger demand opportunity enters a bounded expanded-capacity case with extra factory spending and financing. Company demand statements supply a ceiling on the opportunity; the forecast still requires qualified delivery.

Customers accelerated the shift from 800G toward 1.6T. Q4 DSP and raw-material constraints limit that transition; management expects a material DSP improvement in Q1 2027. Two DSP suppliers are qualified, with a third being qualified. Shared assembly capacity and separate final testing can constrain the generations differently.

Long-term agreements and pricing

The first 1.6T agreement was signed in October with an unnamed existing tier-one customer. Its June 2026 through May 2029 measurement period includes a minimum 10% of the customer's requirements, subject to supply conditions. Management estimates $1-1.5 billion of revenue over three years. Contractual price caps and reductions apply. The agreement contains no committed customer capital funding or prepayment.

A second agreement is in final negotiations. Management describes a potential value at least five times the first and a wider scope including 800G, lasers and other products. Its term, binding volume, detailed price schedule and product split remain undisclosed. The model assigns a 55% signing hypothesis and a 20% conditional ability to qualify and fund incremental capacity, producing an 11% joint expanded outcome. Existing constrained sales absorb overlapping contract demand. The headline contract value receives no separate revenue addition or Base bonus.

3. Business, supercycle and market capture

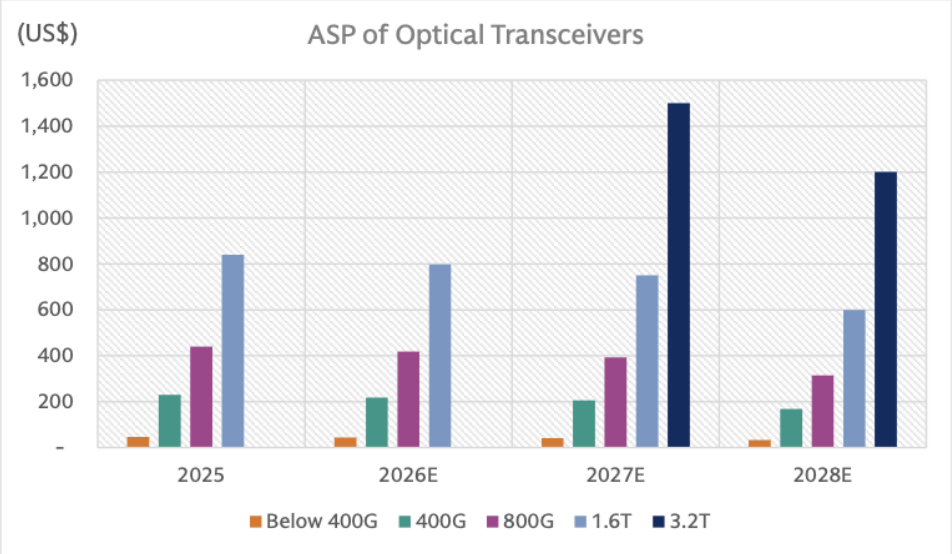
Bandwidth demand rises as training and inference clusters deploy larger fabrics, more optical links and faster interconnects. AAOI sells modules into that deployment cycle and manufactures lasers internally. Internal laser production can improve availability, unit costs and qualification control. Module design wins, accepted output and customer pricing determine the resulting profit.

Framework application phase	Weight	Company capture treatment
AI training / inference clusters and hyperscale networks	100% thematic	800G, 1.6T and separately qualified external light sources
Wider cloud, edge and physical-AI connectivity	50% thematic	Research context; no second overlapping revenue wallet
Further distributed compute / emerging photonics	20% thematic	No unsourced company design wins or revenue

The retained September 2026 broker market summary puts global module/transceiver revenue at $68 billion in 2026, $131 billion in 2027 and $149 billion in 2028. It is a dated broker estimate with broad product coverage. Its reported 2027 unit forecasts are about 49 million 800G, 71 million 1.6T and 23 million 3.2T modules. The issuer provides no comparable companywide market-share series. Company forecasts therefore use the factory and product model, with the broker forecast serving as a demand-scale cross-check.

The retained industry ASP chart suggests approximate annual benchmarks of $400 for 800G and $750 for 1.6T in 2027, falling toward $320 and $600 in 2028. AAOI realized prices can differ by reach, architecture, customer, location and contract. Current net ASP assumptions include price erosion; customer-warrant accounting is already inside recognized net revenue.

Exhibit 4: Global Optical transceivers: Assuming ASP decline YoY for like-for-like products



Source: Company data, Goldman Sachs Global Investment Research

Retained owner-supplied Goldman Exhibit 4: broad industry ASP benchmarks. Original source image and vintage preserved. Model inputs below use approximate chart readings.

Benchmark USD/module	2026E	2027E	2028E
800G	420	400	320
1.6T	800	750	600
3.2T	Not shown	1,500	1,200

The Q4 2027 planned case uses $377 per 800G module and $690 per 1.6T module. The 2028 schedule declines each quarter. A 3.2T case displaces a bounded part of existing 1.6T slots, with lower throughput and higher unit cost. It creates no additional free manufacturing capacity. The model's 2027 revenue remains a small fraction of the retained broad global module forecast; the more demanding checks are the company's accepted output, product mix and financing.

4. Numerical probabilities and overlap

Signal / uncertainty	Probability	Bounded conditional treatment
Current signed LTA	100% event status	Demand visibility within constrained deliveries; agreed pricing
Planned physical ramp	64%	Quarter-average capacity toward 1M/month and late-2027 milestone
Additional delay	25%	Another common quarter of power / qualification delay; costs and funding retained
Expanded and accelerated output	11% joint	55% larger LTA x 20% ability to qualify/fund extra output; extra capex and debt
External ELSFP commercialization	60%	Qualified capacity x explicit acceptance; 40% limited-pilot fallback
Usable cash-tax shield	35%	Finite dated $79M gross NOL-DTA upper-bound proxy
Conditional capital support	50%	Bounded qualifying CHIPS / TSIF cash; separate from valued OP
Retained policy exposure	40%	Filtered external-light-source repricing only
3.2T within policy branch	30% conditional / 12% joint	Substitute 1.6T slots from 2028; 80% throughput, 1.5x cash unit cost

Three physical states, two ELSFP outcomes, two cash-tax outcomes, two support outcomes and three policy outcomes create 72 coherent joint leaves. Each leaf carries the same correlated physical outcome across its quarters. The price is calculated after that leaf's costs, cash, financing and shares. The model then weights those per-share values. Multiplying a weighted earnings total by an independently averaged share count would lose this dependence.

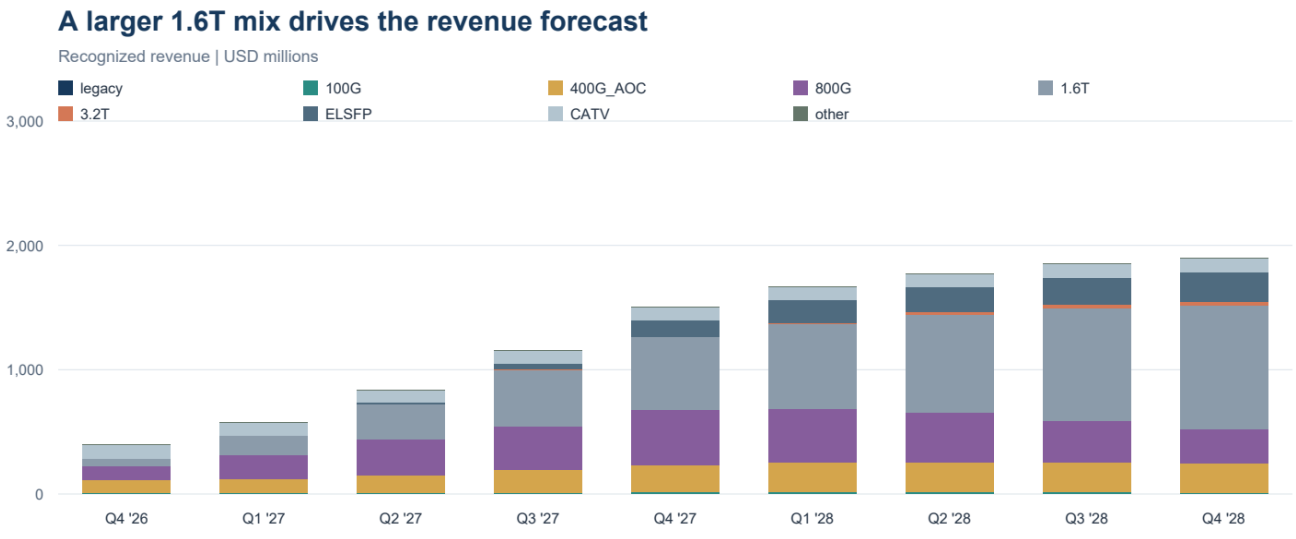
The official ramp, qualification evidence, contract demand and broker shipment revisions share substantial overlap. Recognized sales follow one customer/product/capacity ledger. The physical scenarios capture common timing and acceptance uncertainty across quarters. Internal lasers remain an input to company modules, while external ELSFP sales enter their own customer wallet.

The signed LTA carries no supplier-funding commitment, so it earns no prepayment score bonus. Additional customer funding or bank availability requires a separate financing assumption. The larger contract remains a conditional commercial signal. The model can price that signal before a final announcement while keeping its source gaps and delivery constraints visible.

5. Product revenue and manufacturing economics

Current and future quarters carry Model status. Q4 company totals constrain the first modeled quarter; later product schedules use management's milestone, finite equipment capacity, qualified conversion, product mix and price erosion. Product-level forecasts retain limited confidence because the issuer supplies neither a full SKU shipment schedule nor future product gross margins.

Product	Revenue law	Cost and overlap treatment
Legacy / 100G	Explicit small quarterly schedules	Finite legacy demand; separate cash margins
400G / AOC	Low-speed DC total less legacy and 100G	Mixed-product margin proxy; no generic 800G ASP substitution
800G	Accepted units x net realized ASP	Shared high-speed slots; cash margin plus factory D&A / SBC
1.6T	Accepted units x net realized ASP	Shared slots, distinct final-test requirements
3.2T	Conditional substitution of 1.6T slots	80% output throughput; 1.5x 1.6T cash unit cost
ELSFP	External capacity x conversion x $400 ASP	60% qualification branch; pilot fallback; dedicated cost
CATV	Guidance-constrained near term; finite later run rate	Longer collection terms and customer concentration
Other	Explicit small schedule	Residual business; no unexplained profit plug



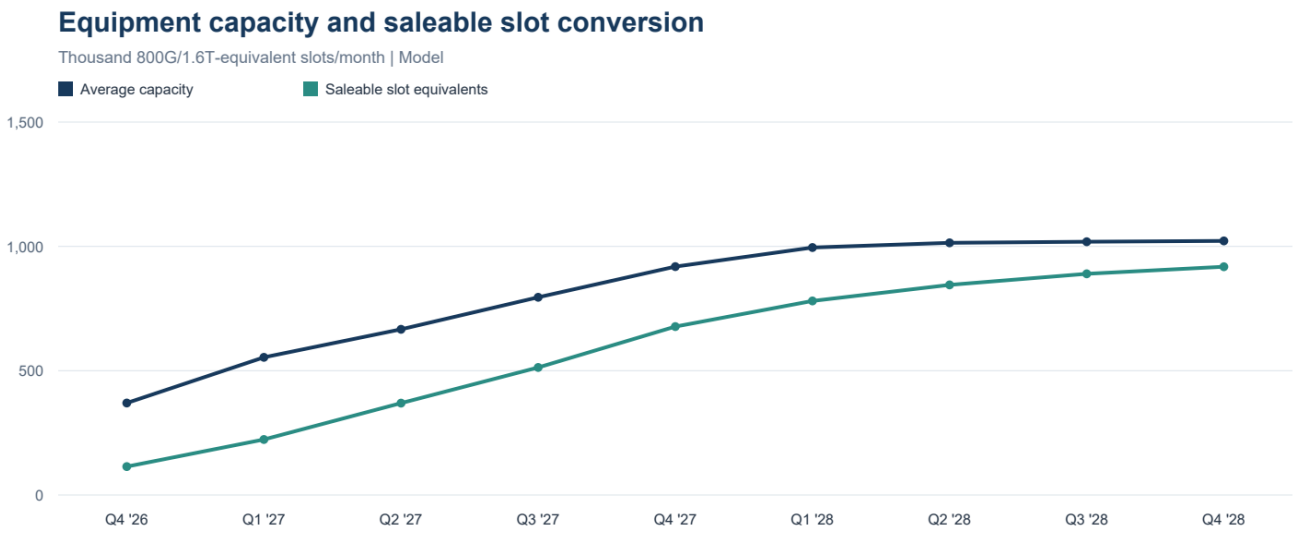
Probability-weighted recognized product revenue. External light sources are separate from lasers consumed in AAOI transceivers.

Q4 2026 through Q4 2027 | revenue, USD M

Product	Q4 2026	Q1 2027	Q2 2027	Q3 2027	Q4 2027
legacy	3	3	3	3	3
100G	10	11	12	12	12
400G_AOC	100	106	134	179	218
800G	112	192	291	355	448
1.6T	60	157	286	453	583
3.2T	0	0	0	0	0
ELSFP	1	3	10	50	132
CATV	110	100	100	105	110
other	3	3	3	3	3
Total	399	575	838	1,160	1,508

Calendar 2028 | revenue, USD M

Product	Q1 2028	Q2 2028	Q3 2028	Q4 2028
legacy	3	3	3	3
100G	12	12	11	11
400G_AOC	235	239	236	232
800G	438	403	339	274
1.6T	679	789	906	994
3.2T	6	15	27	31
ELSFP	191	207	222	238
CATV	105	105	105	110
other	3	3	3	3
Total	1,672	1,775	1,852	1,897



Quarter-average capacity and saleable 800G/1.6T-equivalent production slots. A 3.2T module consumes 1.25 equivalent slots under the modeled throughput assumption. Physical outcomes, product conversion and mix determine slot usage.

Quarter	Capacity K/month	Saleable K slots/month	High-speed slot utilization
Q4 2026	370	114	31%
Q1 2027	554	223	40%
Q2 2027	666	370	55%
Q3 2027	795	513	65%
Q4 2027	919	677	74%
Q1 2028	995	781	78%
Q2 2028	1,014	845	83%
Q3 2028	1,019	890	87%
Q4 2028	1,022	918	90%

Quarter	800G net ASP	1.6T net ASP	800G units K	1.6T units K
Q4 2026	420	800	268	75
Q1 2027	410	780	469	201
Q2 2027	400	750	727	381
Q3 2027	390	720	909	629
Q4 2027	377	690	1,187	845
Q1 2028	340	648	1,288	1,047
Q2 2028	324	618	1,243	1,277

Quarter	800G net ASP	1.6T net ASP	800G units K	1.6T units K
Q3 2028	308	588	1,101	1,540
Q4 2028	293	557	937	1,785

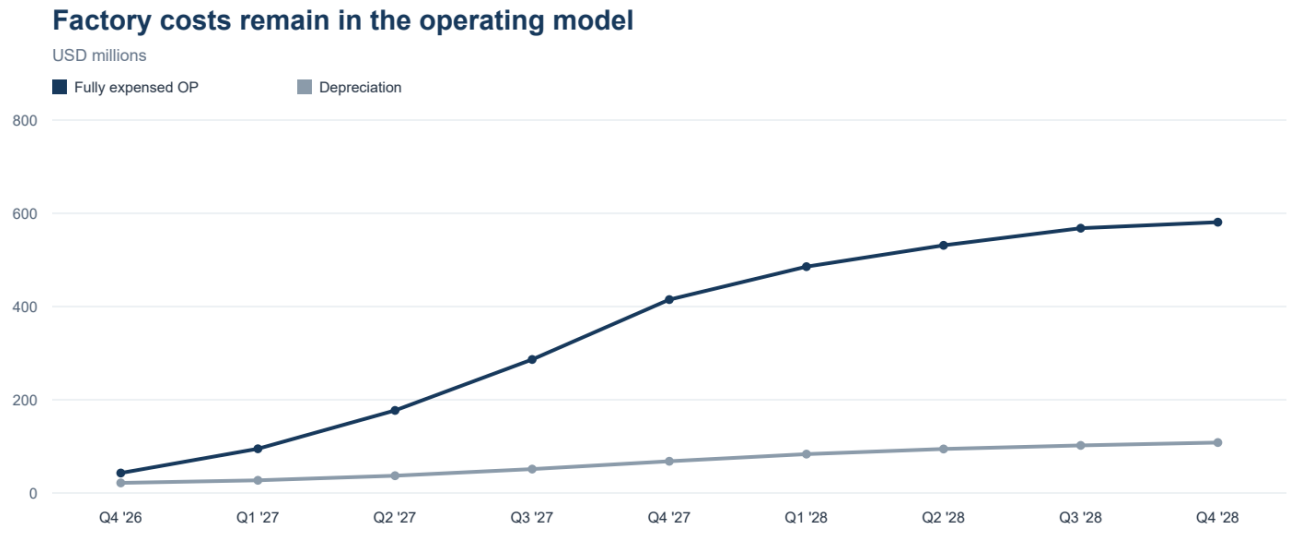
6. Cost, operating-profit and asset bridges

Cash product cost equals revenue times one minus the selected cash contribution margin. New capital payments first distinguish deposits, land and depreciable factory assets. Deposits release into construction in progress when equipment is received; commissioning transfers the available balance into in-service cohorts. Depreciation follows those cohorts with a half-quarter first-period convention. Land remains separate.

The opening bridge uses $797 million of net depreciable in-service assets, $235 million construction in progress and $12 million land. The $466 million capital-deposit opening balance is a scope-qualified proxy from the long-term asset note. Deposit releases are capped by the available opening balance plus new cash deposits. Reclassifying a paid deposit causes no second cash payment.

The modeled remaining life for opening assets is ten years. New commissioned cohorts use 80% machinery at seven years, 15% buildings at twenty-five years and 5% information technology at five years. These are explicit analyst useful-life and asset-mix proxies. The model expenses total depreciation and recurring stock compensation once, allocates factory charges to product GP and adds noncash charges back once in cash flow.

R&D and SG&A have separate quarterly fixed schedules. Incremental revenue above the Q4 calibration carries 2.5% R&D and 4% SG&A support cost. Fixed engineering costs continue through the delayed outcome. Recurring stock compensation rises from $6 million to $10 million per quarter. Intangible amortization remains an expense. Policy increments carry their separate incremental support cost.



Fully expensed operating profit and modeled factory depreciation. Lower output retains the installed-asset cost and fixed support budget.

Quarter	Revenue	GP	D&A	Opex	OP	OP margin
Q4 2026	399	121	22	78	43	11%
Q1 2027	575	187	27	92	95	17%

Quarter	Revenue	GP	D&A	Opex	OP	OP margin
Q2 2027	838	289	37	112	177	21%
Q3 2027	1,160	422	51	135	286	25%
Q4 2027	1,508	575	68	161	415	28%
Q1 2028	1,672	659	83	174	486	29%
Q2 2028	1,775	714	94	183	531	30%
Q3 2028	1,852	758	102	190	568	31%
Q4 2028	1,897	776	108	196	581	31%

Product gross profit | USD M

Product	Q4 2026	Q1 2027	Q2 2027	Q3 2027	Q4 2027
legacy	0	0	0	0	0
100G	2	2	2	2	2
400G_AOC	23	26	35	48	60
800G	33	59	94	117	152
1.6T	23	62	117	189	248
3.2T	0	0	0	0	0
ELSFP	0	1	5	27	74
CATV	38	35	36	37	38
other	1	1	1	1	1
Total	121	187	289	422	575

Product	Q1 2028	Q2 2028	Q3 2028	Q4 2028
legacy	0	0	0	0
100G	2	2	2	2
400G_AOC	65	64	61	60
800G	151	139	116	92
1.6T	294	347	401	433
3.2T	3	8	16	19
ELSFP	108	117	126	135
CATV	35	35	34	36

Product	Q1 2028	Q2 2028	Q3 2028	Q4 2028
other	1	1	1	1
Total	659	714	758	776

Weighted full-year 2027 revenue is $4,081 million and operating profit is $973 million. Calendar 2028 revenue is $7,197 million and OP is $2,166 million. These annual sums cross-check the quarterly build; each valuation retains its own rolling window.

7. External light sources, engineering and supply

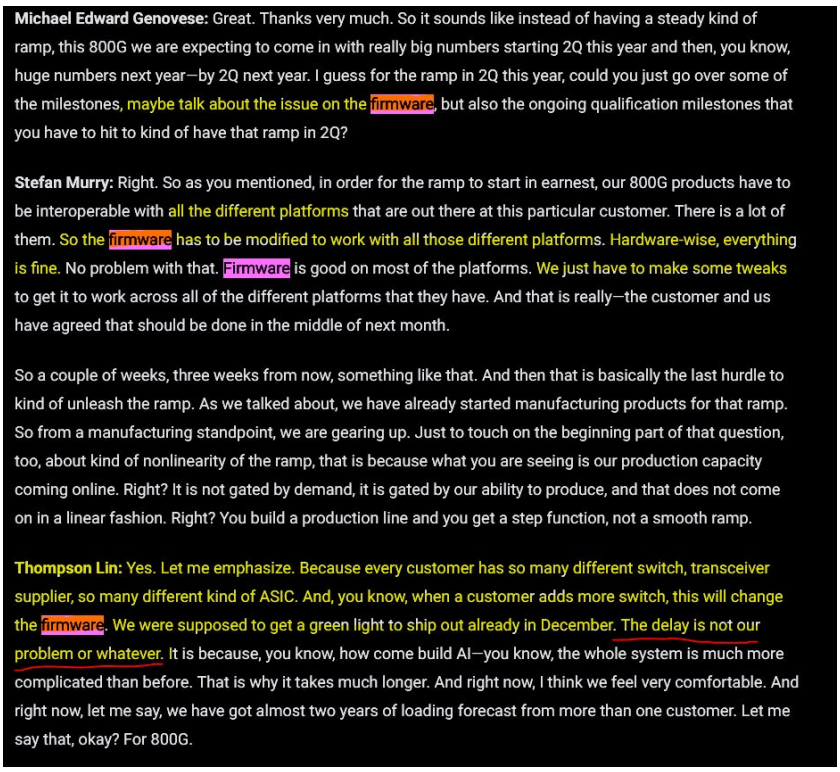
Four or five high-power laser customers are in qualification, with positive preliminary test feedback. Management describes strong bare-laser gross margins and earlier external-module margin potential. Bare lasers and packaged ELSFP modules have different ASPs, packaging costs, yields and qualification requirements. The $400 modeled external-module price is an explicit retained proxy.

New high-power laser tools can take 18-24 months from equipment prepayment to volume production. Prior tool orders began earlier, so that lead time is attached to each investment cohort. It starts again only for a new equipment order. Internal transceiver requirements retain priority before external sales. The official September deck's end-2027 ELSFP capacity milestone and an earlier verbal 2028 milestone have different dates; both are preserved as source context.

Quarter	Nominal ELSFP K/month	Qualified conversion	Expected external revenue, USD M
Q4 2026	5	10%	1
Q1 2027	20	15%	3
Q2 2027	50	25%	10
Q3 2027	150	45%	50
Q4 2027	300	60%	132
Q1 2028	400	65%	191
Q2 2028	400	70%	207
Q3 2028	400	75%	222
Q4 2028	400	80%	238

A qualified branch uses module acceptance rising from 10% near term to 80% late in the forecast. A 40% fallback keeps only limited pilot sales. The policy premium reaches 5% on 50% eligible exposure and 60% repricable exposure, phased over time. This narrow filter respects the first LTA's price caps and avoids applying a general premium to all company transceivers.

InP supply is described as adequate through 2027, supported by six suppliers and a 2028 expansion agreement. The new Chinese supply agreement permits up to $194 million of annual purchases and $40 million prepayment, with export-approval and cancellation/refund conditions. The model funds $20 million in Q4 and $20 million in Q1 2027. Component purchases remain inside unit COGS. The maximum purchase amount receives no second expense or revenue addition.



Michael Edward Genovese: Great. Thanks very much. So it sounds like instead of having a steady kind of ramp, this 800G we are expecting to come in with really big numbers starting 2Q this year and then, you know, huge numbers next year—by 2Q next year. I guess for the ramp in 2Q this year, could you just go over some of the milestones, maybe talk about the issue on the firmware, but also the ongoing qualification milestones that you have to hit to kind of have that ramp in 2Q?

Stefan Murry: Right. So as you mentioned, in order for the ramp to start in earnest, our 800G products have to be interoperable with all the different platforms that are out there at this particular customer. There is a lot of them. So the firmware has to be modified to work with all those different platforms. Hardware-wise, everything is fine. No problem with that. Firmware is good on most of the platforms. We just have to make some tweaks to get it to work across all of the different platforms that they have. And that is really—the customer and us have agreed that should be done in the middle of next month.

So a couple of weeks, three weeks from now, something like that. And then that is basically the last hurdle to kind of unleash the ramp. As we talked about, we have already started manufacturing products for that ramp. So from a manufacturing standpoint, we are gearing up. Just to touch on the beginning part of that question, too, about kind of nonlinearity of the ramp, that is because what you are seeing is our production capacity coming online. Right? It is not gated by demand, it is gated by our ability to produce, and that does not come on in a linear fashion. Right? You build a production line and you get a step function, not a smooth ramp.

Thompson Lin: Yes. Let me emphasize. Because every customer has so many different switch, transceiver supplier, so many different kind of ASIC. And, you know, when a customer adds more switch, this will change the firmware. We were supposed to get a green light to ship out already in December. The delay is not our problem or whatever. It is because, you know, how come build AI—you know, the whole system is much more complicated than before. That is why it takes much longer. And right now, I think we feel very comfortable. And right now, let me say, we have got almost two years of loading forecast from more than one customer. Let me say that, okay? For 800G.

Historical engineering excerpt retained from the March 2026 owner-supplied research bundle. Customer interoperability and qualified output have separate gates; the current ramp uses the November results and calls.

Dated customer acceptance, factory repeatability and recognized revenue determine commercial conversion. Historical reliability or firmware work supports the learning process; each new generation still requires the customer platform, thermal, optical-power and final-test checks appropriate to that product.

8. Working capital, capital payments and cash

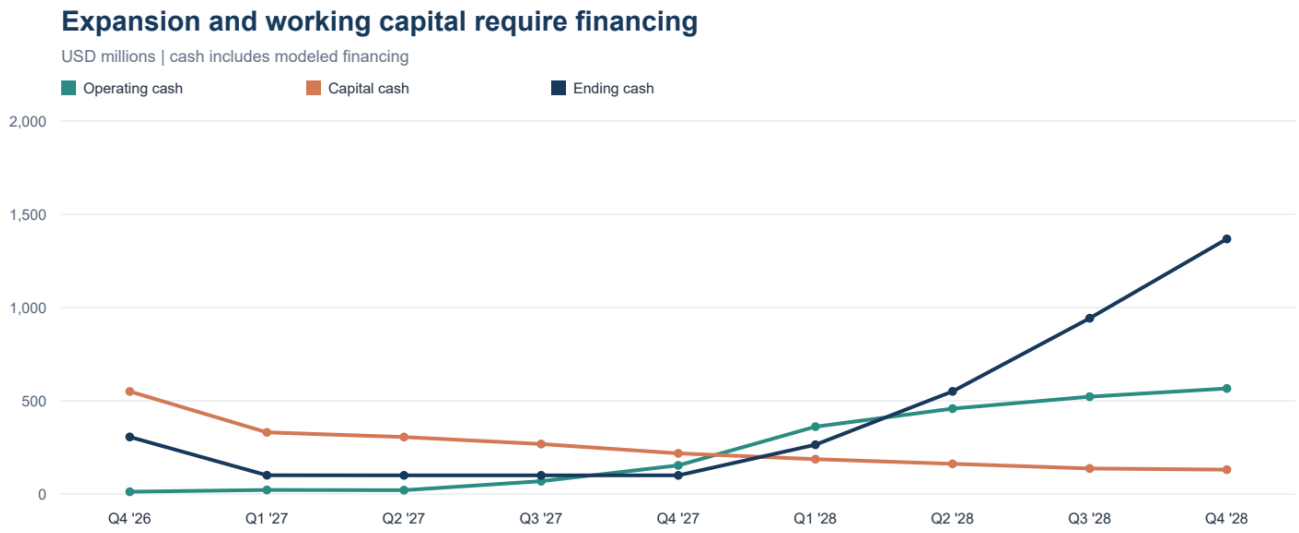
September unrestricted cash was $605 million, with a further $16 million restricted. Accounts receivable were $414 million and inventories $378 million. Digicomm accounted for approximately $235 million of receivables under extended payment terms. Accounts payable were $494 million. The filing discloses a change in capital-related payables, without an ending capital-payables balance.

The forecast separates a $60 million estimated capital-payables balance from operating trade payables. Datacenter collections use 100 days, CATV 190 days. Inventory days decline from 140 toward 100, while the cost-based trade-payable proxy declines from 210 toward 180 days. These are analyst cash-cycle assumptions calibrated to the unusually large disclosed receivable, inventory and payable balances. A shortening of supplier credit or a delay in customer cash collections raises external funding needs materially.

Nine-month operating cash use was $77 million, cash PP&E payments were $702 million and equipment/other prepayments were $407 million. Q3 capital investments were $501 million, including $143 million of prepayments. Management expects Q4 capital payments above Q3. The model uses $550 million in Q4 and separate 2027 payment schedules by physical state. Timing of capital cash, paid deposits, asset commissioning and production capacity remains explicit.

Quarter	CFO	Capital cash	FCF	Receivables	Ending cash
Q4 2026	12	550	-538	557	306
Q1 2027	22	331	-309	745	101

Quarter	CFO	Capital cash	FCF	Receivables	Ending cash
Q2 2027	20	305	-285	1,041	100
Q3 2027	69	268	-199	1,408	100
Q4 2027	153	218	-65	1,805	100
Q1 2028	361	187	174	1,984	264
Q2 2028	458	162	296	2,100	550
Q3 2028	522	137	385	2,187	943
Q4 2028	566	130	435	2,242	1,368



Capital payments include new equipment deposits. Ending cash includes the modeled financing schedule and conditional support receipts. FCF is operating cash flow less capital cash payments.

Capital commitments of approximately $929 million and the remaining construction obligations require careful overlap checks; the model receives no automatic revenue credit from purchase commitments. Purchase options on further property remain unexercised and receive no automatic cash-outlay assumption. Existing loan renewal, collateral eligibility and supplier terms remain financing conditions.

9. Debt, shares and funding requirements

The October EWB package comprises a $170 million term facility and $30 million revolver. It replaces the old unused $35 million BOKF line. The model retains a further $28 million of other unused capacity after removing that replaced line. Term borrowing uses a forecast 6.85% rate and approximately $6 million quarterly principal amortization from Q1 2027. The first amortization date is a conservative proxy because the full contract was unavailable.

Existing Asian borrowings of $97 million, bank acceptances of $52 million and $125 million convertible principal remain separately funded. The convertible coupon uses the contractual 2.75% rate. The source's stated quarterly interest amount and EPS-note prose have inconsistencies with the principal/coupon and reported table. The report retains reported actuals, uses the contractual forecast coupon and keeps gross economic conversion shares in the valuation denominator.

The delayed state assumes a finite, uncommitted $50 million bank expansion in Q3 2027. The expanded-capacity state assumes $200 million of uncommitted additional borrowing. The selected model also assumes future equity can be raised at $45 per share with 2% fees. Issuance follows the cash shortfall to a $100 million model buffer after available debt. The cash buffer is an analyst control; loan covenant minima were unavailable.

The completed third ATM raised about $588 million net across the full program. Roughly $514 million was already inside September financing cash. Only the approximately $74 million residual after September enters the current cash bridge. This treatment prevents the completed financing from being counted twice. The later legal count was 90,770,732 common shares on 2 November, compared with 89,885,000 at September end.

Economic reservation	Million shares	Treatment
September legal common	89.885	Beginning-Q4 anchor
Full Amazon warrant reservation	7.945	Gross reserve; no hypothetical cash proceeds
Convertible if-converted shares	2.884	Included once; coupon retained conservatively
RSUs / maximum PSU payout	1.087 / 1.258	Full reserved-right allowance
Starting Q4 economic shares	103.060	Sum of source-anchored beginning shares / rights
Current authorized common	120.000	Binding modeled gross reservation cap

Future baseline awards add a small reserved-share allowance. New external equity raises add shares only after issuance, so each ceiling uses the stated quarter's beginning economic count. Already reserved warrants, notes and awards receive no second dilution layer. Customer-warrant revenue accounting remains a separate noncash revenue adjustment. Net ASPs already include the contra-revenue effect; cash flow adds the noncash charge back once.

Quarter	New equity shares, M	Cumulative, M	Starting economic shares, M	Debt, USD M
Q4 2026	0	0	103	443
Q1 2027	1	1	104	483
Q2 2027	6	7	106	511
Q3 2027	4	11	112	518
Q4 2027	2	13	116	512
Q1 2028	0	13	118	506
Q2 2028	0	13	118	499
Q3 2028	0	13	118	493
Q4 2028	0	13	118	487

Selected state-specific cumulative equity issuance ranges from roughly 9 million to 15 million shares. Minimum gross share-authorization headroom is only about 147,000 shares in the most demanding selected case. The forecast is highly sensitive to financing price, payment terms and spending. Additional capital, expanded

authorization or a slower factory plan could be necessary if the funding assumptions fail. The failed stresses below retain that gap explicitly.

10. Taxes, grants and one-time accounting

Normalized book tax is 21% on positive recurring pretax income. The source maintains a full valuation allowance on US and Taiwan deferred-tax assets. A future accounting release can change GAAP net income while contributing no operating profit or incremental cash at that moment. It is excluded from the valued earnings chain.

The cash-tax branch assigns a 35% chance to using a finite $79 million benefit cap derived from the older gross NOL-related deferred-tax amount. Jurisdiction, taxable-income mix, restrictions and the 2026 attribute roll-forward remain unverified. The conditional branch offsets at most 80% of normalized quarterly tax until its finite balance is exhausted. The remaining branch pays normalized cash tax.

The capital-support branch assigns 50% probability to bounded eligible CHIPS and TSIF cash receipts. The IRS source describes a 35% advanced-manufacturing investment credit for qualifying property placed in service after 2025, subject to construction beginning before 2027. The model applies explicit US, semiconductor-eligibility and in-service fractions, with a narrower grandfathered 2027 cohort. Company-specific qualification and claim timing remain assumptions.

The TSIF award has a $21 million ceiling through December 2027. Approximately $4 million was recognized in Q3 and sits in a receivable. That existing receivable is separated from the remaining conditional award. Cash receipts enter their modeled quarters. Grants, refunds and a one-time deferred-tax release contribute no premium to operating profit or recurring revenue. A no-support calculation leaves the selected operating forecast unchanged.

11. Analytical scorecard and counter-evidence

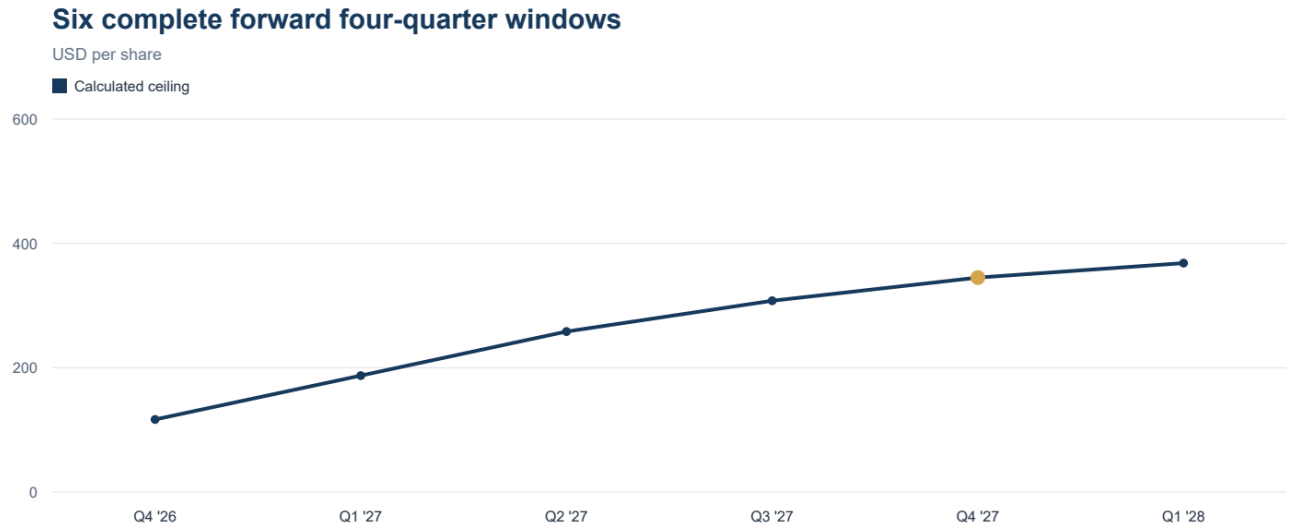
Category	Score	Current evidence and limitation
Core thesis	14/20	Internal lasers, design and manufacturing support capture; competitive standardized modules constrain returns
Cycle	9/10	High-speed ramp is commercial; new power and component timing constrain near-term conversion
TAM expansion	8/10	AI bandwidth and external light sources expand applications; company wallet overlap is removed
Catalysts	5/5	Accepted 1.6T shipments, Texas power/startup, customer LTAs and qualified external light sources
Macro	9/10	AI networks require bandwidth and optical links; deployment/capex timing remains variable
Alpha	2/5	Forecast requires a demanding output/cost ramp and large external funding
Risk deductions	(5)	Five existing risk classes, one point each
Total	42/60 = 70	Protected Excel Base 72 remains in place

The five risk classes are execution and components; competition and customer concentration; external-light-source qualification; funding and dilution; and architecture/legacy-demand transition. The new results sharpen those same risks. The signed contract improves visibility, while price caps, revised timing and financing needs limit the score benefit. No extra score deduction is layered on top of a physical delay already represented in the shared scenario.

Required evidence	Economic consequence	Reassessment trigger
Accepted 800G / 1.6T output by quarter	Units, mix, price, unit cost and acceptance timing	Missed accepted-output ramp or weaker realized ASP
Texas power and installed final-test throughput	Commissioning, depreciation and capacity availability	Further delay or output below installed equipment
Executed second LTA and SKU pricing	Bounded customer allocation and contract contribution	Term/pricing incompatible with assumed share or margin
ELSFP qualification and volume orders	External wallet, yield, packaging and margin	Pilot status persists or qualification scope changes
Customer collections / supplier terms	Cash conversion and equity funding	Longer DSO, shorter payable terms or deferred collection
Funded credit and actual share issuance	Cash, interest, authorization and per-share ceiling	Unfunded obligations or economic shares above reservation

12. Current rolling valuation and sensitivity

Each of the 72 coherent states supplies current-plus-next-three quarterly fully expensed operating profit, multiplied by 20 and divided by that state's starting-quarter economic shares. The state-level per-share results are probability weighted. Currency is USD and FX is one. The established company convention uses operating profit directly as the valuation proxy. Debt, interest, cash and dilution remain explicit financial diagnostics under that retained convention.



Current modeled ceilings. The gold point marks Q4 2027. All six current windows use state-specific earnings and funded share counts before probability weighting.

Ceiling quarter	Earnings window	F4Q OP, USD M	Start shares, M	Ceiling, USD
Q4 2026	Q4 2026 to Q3 2027	601	103	117
Q1 2027	Q1 2027 to Q4 2027	973	104	187
Q2 2027	Q2 2027 to Q1 2028	1,364	106	258

Ceiling quarter	Earnings window	F4Q OP, USD M	Start shares, M	Ceiling, USD
Q3 2027	Q3 2027 to Q2 2028	1,718	112	308
Q4 2027	Q4 2027 to Q3 2028	2,000	116	345
Q1 2028	Q1 2028 to Q4 2028	2,166	118	368

The weighted earnings and average-share columns aid reconciliation. Exact price values are weighted after division within each state, so dividing the two displayed weighted averages can differ slightly from the selected ceiling. The operating forecast ends Q4 2028. Q1 2028 is the last complete rolling valuation window; later price-target cells preserve their existing n.a. semantics.

Funding sensitivities | separate unselected calculations

Changed input	Maximum capital gap at share cap, USD M	Selected weight
equity price 40	67	0%
uncommitted debt unavailable	90	0%
capex 20pct higher	330	0%
accepted highspeed 20pct lower	24	0%
cash support unavailable	0	0%
DC collections 30 days later	502	0%
trade terms 30 days shorter	293	0%

The default forecast assumes the finite debt and equity funding described above. A lower equity issuance price, unavailable conditional debt, higher capex or weaker cash terms can create obligations that exceed current share authorization. Those calculations retain the capital gap. They supply no funded valuation. Future confirmed financing, changed authorization or a source-supported slower operating plan would require a coherent new company model.

13. Source record and completeness

Primary results: AAOI Q3 2026 earnings release and financial tables; original SEC exhibit and issuer mirror captured. Statements: Q3 2026 Form 10-Q, including balance sheet, operating statement, cash flow, debt, awards, tax, commitments and subsequent events. Management: complete Q3 and Q2 call transcripts, prepared remarks and Q&A through operator close, supplied by Quartr through TradingView. Audio was not independently listened to.

The full credit-contract and annual-report original download attempts returned access errors. Indexed primary loan and tax excerpts support the qualified inputs. Exact covenant tests, the first term-amortization date, the 2026 tax-attribute roll-forward and company CHIPS eligibility remain gaps. The reported EPS-note/table inconsistency and convertible-interest amount are retained as source limitations. Conservative modeled coupon and gross-reservation conventions are stated above.

Retained secondary context includes the dated owner-supplied Goldman ASP exhibit and March engineering article excerpt. Historical anonymous analyst notes and the InP expert-call originals remain recoverable with their source

identities and prior decision records. Their overlapping timing and cost hypotheses are reconciled in the current common scenarios. They supply no automatic second module sale, price premium or production haircut.

Source	Locator / publication
Q3 Form 10-Q	https://appliedoptoelectronics.gcs-web.com/node/17736/html
Q3 earnings release	SEC 000168316826008351 / aaoi_ex9901.htm
Q3 full call	Quartr 4480007 | 5 November 2026
Q2 full call	Quartr 4015494 | 6 August 2026
EWB financing	SEC 000168316826008310 / aaoi_8k.htm and indexed primary agreement excerpts
Advanced manufacturing credit	irs.gov/credits-deductions/advanced-manufacturing-investment-credit
Broker / engineering context	Retained September 2026 Goldman Exhibit 4 and March 2026 owner research bundle

Financial amounts in the display are rounded; exact inputs and outputs remain in the linked model and standalone formula workbook. The model contains numerical probabilities and bounded conditional outcomes, a reconciled product/cost build, finite assets and cash-tax balances, explicit debt/equity funding, protected share conventions and six complete rolling windows. Published earlier-quarter forecasts retain their original values and vintage.